

## PART I · FOUNDATIONS

### Physics, Silicon & Hierarchy

#### Ch 1: The Boundary

The Physical AI Imperative · Causal Loop

#### Ch 2: Constraints

5 Physical Columns · Latency Freshness Wall

#### Ch 3: Cognition

5 Cognitive Stages · The Great Systems Tension

#### Ch 4: Hierarchy

Multi-Rate Topology · Three Cadences

Formulate

## PART II · THE 5 ORGANS

### Sensory-Motor Execution Pipeline

#### Ch 5: Perception

Sensory Transduction · Spatial Tokens

#### Ch 6: Memory & State

SE(3) World Models · Frame Trees

#### Ch 7: Reasoning & Intent

VLM Grounding · Expiring Intent Leases

#### Ch 8: Planning & Synthesis

Diffusion / ACT ·  $C^2$  Jerk Continuity

#### Ch 9: Reflex Enforcement

1 kHz MCU Enforcer · CBF Safe Sets

Deploy

## PART III · RELEASE

### Hardware, Governance & Release

#### Ch 10: Placement

4-Tier Placement · Thermal Budgets

#### Ch 11: Governance

Authority Arbitration · Bumpless Override

#### Ch 12: Assurance

CAE Safety Cases · 4-Rung Qualification

#### Ch 13: The Frontier

Dual-Brain Kit · Oral Defense Dossier